Hydrop enables droplet-based single-cell ATAC-seq and single-cell RNA-seq using dissolvable hydrogel beads

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Abstract Single-cell RNA-seq and single-cell assay for transposase-accessible chromatin (ATAC-seq) technologies are used extensively to create cell type atlases for a wide range of organisms, tissues, and disease processes. To increase the scale of these atlases, lower the cost and pave the way for more specialized multiome assays, custom droplet microfluidics may provide solutions complementary to commercial setups. We developed HyDrop, a flexible and open-source droplet microfluidic platform encompassing three protocols. The first protocol involves creating dissolvable hydrogel beads with custom oligos that can be released in the droplets. In the second protocol, we demonstrate the use of these beads for HyDrop-ATAC, a low-cost noncommercial scATAC-seq protocol in droplets. After validating HyDrop-ATAC, we applied it to flash-frozen mouse cortex and generated 7996 high-quality single-cell chromatin accessibility profiles in a single run. In the third protocol, we adapt both the reaction chemistry and the capture sequence of the barcoded hydrogel bead to capture mRNA, and demonstrate a significant improvement in throughput and sensitivity compared to previous open-source droplet-based scRNA-seq assays (Drop-seq and inDrop). Similarly, we applied HyDrop-RNA to flash-frozen mouse cortex and generated 9508 single-cell transcriptomes closely matching reference single-cell gene expression data. Finally, we leveraged HyDrop-RNA's high capture rate to analyze a small population of fluorescence-activated cell sorted neurons from the *Drosophila* brain, confirming the protocol's applicability to low input samples and small cells. HyDrop is currently capable of generating single-cell data in high throughput and at a reduced cost compared to commercial methods, and we envision that HyDrop can be further developed to be compatible with novel (multi) omics protocols.

Editor's evaluation

De Rop et al. introduce a flexible microfluidics-based single-cell genomics technology that expands and improves previously existing custom droplet-based scRNA-seq protocols (inDrops and Drop-seq) in interesting directions: better data quality, simplified workflow, high-cell recovery, and flexibility towards other single-cell applications.

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